

Bachelor AI - MLOps

01 SCHEDULE AND ORGANIZATION

Tobias Mérinat

`tobias.merinat@hslu.ch`



DEPARTMENT OF INFORMATION TECHNOLOGY
Lucerne University of Applied Sciences and Arts
6343 Rotkreuz, Switzerland

September 17/24 and October 15 2026

Fundamentals of MLOps

Bringing Machine Learning Models to Production

- Topics
 - Challenges, Experiment Tracking, Model Registry, Processing and Prediction Modes, Feature Types, Batch Inference, Stream Processing, FTI Architecture, Monitoring, Data Drift, Feature Stores
- Lots of alternation between theory and practice:
 - 7 theory blocks
 - 9 guided exercises
 - One block with free-form exercises

- 01 Schedule and Organization <- **you are here**
- 02 About Me
- 03 Warmup Discussion
- 04 MLOps Definition and Challenges
- 05a-d: Exercise Infrastructure, Codespaces and Exercise Kickstart

- 06 **Exercise** Introduction to the Exercise Infrastructure
- 07 **Exercise** A Simple Machine Learning Model
- 08 Model Platforms, Registry, Experiment Tracking, MLFlow
- 09 **Exercise** Setting up Model Registry Service
- 10 **Exercise** A Simple Model with MLFlow

- 11 Processing and Prediction Modes, Features
- 12 **Exercise** Batch Inference Pipeline

- 13 Stream Processing
- 14 **Exercise** Stream Processing Infrastructure
- 15 FTI Pipeline Architecture
- 16 **Exercise** Streaming Inference Pipeline

- 17 Data Drift
- 18 **Exercise** Monitoring Infrastructure
- 19 **Exercise** Drift Detection Pipeline

- 20 Feature Stores
- 21 Free-form Exercises